2 *Waves*

2.1 *What is a wave?*

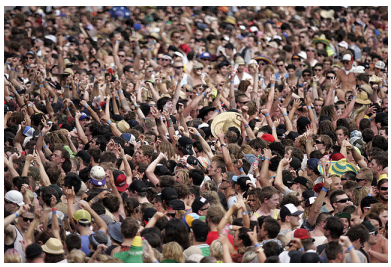


Figure 2.1: A stadium wave.

In a stadium wave, there is one simple instruction: Do what your neighbour (let's say to the left) does, with a slight time delay. Someone starts by jumping up and down, and a column of folks jumping up and down goes round the stadium.

But what is actually travelling around the stadium? Not people; everyone is just jumping up and down in place. It's something quite abstract: It's the *state of being in the air with arms raised*.

Let's analyze a simple model of this situation: Imagine a chain of oscillators, small masses distributed along the x axis, separated by a distance δx. The mass at the left end of the chain, which we will call $x = 0$, jumps up and down, so its height y is a function of time $f(t)$. The next mass, at $x = \delta x$, will reach the same height at a later time $t + \delta t$. So the next mass replicates the behaviour of the first one, with a time delay δt. We can say that the state of the mass at $x = 0$ has travelled a distance δx along the chain, in a time δt.

This repeats for all the masses along the chain, so that the state of the first one, defined as its height as a function of time, propagates down the chain with a velocity $v = \delta x / \delta t$.

Imagine now that you are an observer travelling parallel to the chain of oscillators, at the same velocity, starting from $x = 0$ at $t = 0$. So your position along the x axis as a function of time is $x_{obs} = vt$. Then the position of an oscillator *relative to you* is $x - x_{obs} = x - vt$. Let's call this z.

But since you are moving with the same velocity as the wave, it seems stationary to you! You see the wave frozen in time, as it were. For you the height of any oscillator is a function of position only:

$$y(z) = f(z) \tag{2.1}$$

That means that for someone who is not moving with the wave, the height of an oscillator as a function of its coordinate x and time t is given by

$$y(x,t) = f(z) = f(x - vt) \tag{2.2}$$

What if the instruction for each mass was to copy its neighbour to the *right*? It is not hard to see that in this case we would have a wave travelling to the left, and it would have the form

$$y(x,t) = g(x+vt) \tag{2.3}$$

In general, we can have a superposition of two travelling waves,

$$y(x,t) = f(x-vt) + g(x+vt) \tag{2.4}$$

So, we are now able to state with precision what exactly it is that makes a function of space and time, $y(x,t)$, a wave. It is that x and t are not independent variables in y. If $y(x,t)$ is a wave, then it is in fact a function of a single variable which is a combination of x and t: $z = x \pm vt$.[1] We call z the **phase** of the wave.

[1] The sign of vt depends of the direction in which the wave travels.

Examples of wave motion

In nature, of course, oscillators in a chain do not respond to instructions! Rather, they are linked by mutual forces, but the effect is the same: Each oscillator copies its neighbour, with a slight time delay.

Until now, we have spoken of a *chain* of oscillators, implying a one-dimensional structure, like an elastic string for example. But waves may also propagate in two dimensions (e.g. on a surface of water), or three dimensions (as in a sound wave propagating throughout a room). We won't talk about wave propagation in more than one dimension until later; it is easier to develop the mathematical tools we need if we stick to the simpler one-dimensional case for now.

Some examples of one-dimensional wave motion are:

- a wave in an elastic string;
- a wave in a slinky; or
- a sound wave in a thin column of air.

As you can see from the images in Figs. 2.2–2.4, there are two different types of wave, called **transverse** and **longitudinal** waves. In the first case, the perturbation is in the direction transverse to the direction in which the wave travels (Figs. 2.2 and 2.3). In the other, the perturbation is in the same direction as the propagation of the wave (Fig. 2.4).



Figure 2.2: Waves in guitar strings (Balthus Van Tassel).



Figure 2.3: A transverse wave in a slinky (Nic McPhee).

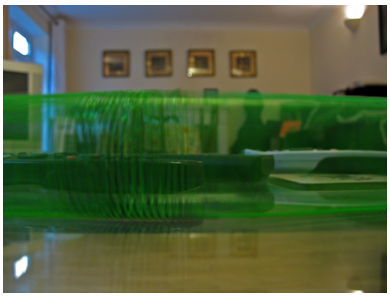


Figure 2.4: A compression wave in a slinky (Zedseayou).

Visualizing a wave

Since a wave is a spatial pattern that changes with time, the best way to capture a wave would be by means of a movie. If we are restricted to still images, we can only visualize one dependence at a time.

A **snapshot** is an image of the wave at a particular instant in time. The images on this page are all snapshots. Mathematically, if the wave is represented by a function $y(x,t)$, and we fix the time $t = t_{fix}$, then $y(x,t_{fix})$ is a function of x only.

If instead we choose to observe the wave as it passes through a fixed position x_{fix}, then $y(x_{fix},t)$ is a function of time only. In that case, we are looking at the motion of a single oscillator in the chain.

A third way of visualizing a wave as a still image is to represent it as a function of the single variable $z = x - vt$. It is what we would see if we were travelling alongside the wave, with the same velocity.

Depending on the circumstances, it may be more convenient to use one or another of the three possible visualizations.

The variable y can represent different things depending on the specific instance of a wave. In the case of a wave in a guitar string, y is the transverse displacement of a tiny segment of the string. In the case of a slinky, it can be the longitudinal displacement of a single coil in the slinky. Or it can be the longitudinal displacement of a thin layer in an air column. In all cases, y measures a displacement from an equilibrium state, for which we can conveniently set $y = 0$. So for example when a stretched string is in equilibrium, it is perfectly straight, and each segment of the string has zero transverse displacement from the equilibrium position. In general, we will call y the **perturbation** (away from equilibrium), and we will call an element of the medium in which the wave propagates (a tiny segment of a string, a coil of a slinky, etc.) a **particle**.

Pulses and harmonic waves

In principle, $y(x - vt)$ can be any function. In these notes, however, we will concentrate on two different families of functions. A **pulse** is a function limited in time. For example, the function

$$y(z) = 0 \text{ if } -w/2 \leq z \leq +w/2 \tag{2.5a}$$

$$y(z) = A\left(1 - \frac{z^2}{(w/2)^2}\right) \text{ otherwise} \tag{2.5b}$$

is a parabolic pulse of height A and width w.

Another pulse wave form frequently seen is the **gaussian pulse**, defined as

$$y(z) = Ae^{-(z/w)^2} \tag{2.6}$$

Strictly speaking, this function is never quite equal to zero, so it shouldn't qualify as a pulse, but it decays rapidly towards zero beyond a few w's on either side of the origin, so it can be treated as a pulse of finite width to a good approximation.

A **harmonic wave** is of the form

$$y(z) = A\cos(kz + \phi_0) \tag{2.7}$$

where A, k, and ϕ_0 are constants. We shall have a great deal to say about harmonic waves later, but for the moment we will continue to explore the properties of a generic wave using a pulse as an example.

2.2 *More mathematical properties of a wave: Partial derivatives*

In this section, we look at the first and second partial derivatives of a wave function $y(x - vt)$ with respect to position and time, and their physical significance.

First, the time derivatives. Since $y(x - vt)$ is the displacement from equilibrium of the particle at position x, the partial derivative $\partial y/\partial t$ is in fact the *velocity* with which the particle at position x moves, as the wave passes by. *Notice that this is quite unrelated to the propagation velocity of the wave itself.*

Similarly, the second partial derivative with respect to time, $\partial^2 y/\partial t^2$, which is calculated at a fixed x, is the acceleration of the particle at that position.

The spatial derivatives are a different matter. $\partial y/\partial x$ is clearly the slope of the function at position x. The second derivative, $\partial^2 y/\partial x^2$, on the other hand, has a more subtle meaning. Figure 2.5 shows the pulse described by the function

$$y(z) = ze^{-z^2} \tag{2.8}$$

where $z = x - vt$.[2] To calculate the derivatives with respect to x, we need to apply the chain rule:

[2] z is assumed to be measured in some units, e.g. cm, so that z is in fact a dimensionless number.

$$\frac{\partial y}{\partial x} = \frac{dy}{dz}\frac{\partial z}{\partial x} = \frac{dy}{dz} \tag{2.9a}$$

$$\frac{\partial^2 y}{\partial x^2} = \left(\frac{d}{dz}\frac{\partial y}{\partial x}\right)\frac{\partial z}{\partial x} = \frac{d^2 y}{dz^2} \tag{2.9b}$$

We are interested in the second derivative, which is explicitly

$$\frac{\partial^2 y}{\partial x^2} = 2z(2z - 3)e^{-z^2} \tag{2.10}$$

In Figure 2.5 the pulse has been plotted highlighting each point according to the magnitude of the second derivative: The higher the value, the darker the point. It is apparent that the highlighting follows the curvature of the graph: The dark portions correspond to a high curvature, and the lightest portions are where the graph is straight. So we conclude that $\partial^2 y/\partial x^2$ is in fact a measure of the curvature of the function $y(z)$.

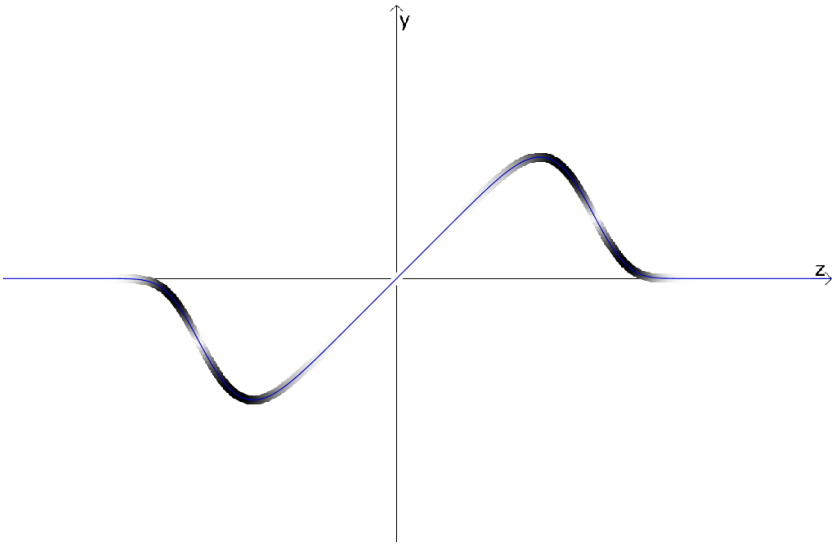


Figure 2.5: A pulse, with emphasis on the sections with a large value of the second derivative.

2.3 *The wave equation*

Let's remember once again that a (one-dimensional) wave is described mathematically by a function of space and time, $y(x,t)$, where the variables x and t are not independent, but in fact appear as the combination $z = x \pm vt$. This ensures that the wave travels along the x axis in one direction or the other, with speed v.

Because x and t are not independent, the partial derivatives of y with respect to x and t are also related. Take the first derivatives, for example:

$$\frac{\partial y}{\partial x} = \frac{dy}{dz}\frac{\partial z}{\partial x} = \frac{dy}{dz} \tag{2.11a}$$

$$\frac{\partial y}{\partial t} = \frac{dy}{dz}\frac{\partial z}{\partial t} = \pm v\frac{dy}{dz} \tag{2.11b}$$

so that

$$\frac{\partial y}{\partial x} = \pm\frac{1}{v}\frac{\partial y}{\partial t} \tag{2.12}$$

This is still not quite a general differential equation that describes any wave, because it differs depending on the direction in which the wave is travelling. But if we differentiate once again, we will get a unique differential equation.

$$\frac{\partial^2 y}{\partial x^2} = \left(\frac{d}{dz}\frac{\partial y}{\partial x}\right)\frac{\partial z}{\partial x} = \frac{d^2 y}{dz^2} \tag{2.13a}$$

$$\frac{\partial^2 y}{\partial t^2} = \left(\frac{d}{dz}\frac{\partial y}{\partial t}\right)\frac{\partial z}{\partial t} = \tag{2.13b}$$

$$= \left(\pm v\frac{d^2 y}{dz^2}\right)(\pm v) = v^2\frac{d^2 y}{dz^2} \tag{2.13c}$$

Now we can say that

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2} \tag{2.14}$$

This is known simply as **the wave equation**. It is a partial differential equation that describes all one-dimensional waves. Its solutions are functions of x and t that propagate along x with speed v.

We won't go further into the wave equation, but just point out one interesting fact about it: Remembering what the second derivatives represent, we see that *at any point x, the instantaneous acceleration of the particle at that point is proportional to the curvature of the wave.*

Example 2.1: *For a gaussian pulse of the form of equation (2.6), calculate the four partial derivatives in terms of $z = x - vt$.*

The particle velocity is

$$\frac{\partial y}{\partial t} = \frac{dy}{dz}\frac{\partial z}{\partial t} = -v\frac{dy}{dz} = \frac{2\,Avze^{\left(-\frac{z^2}{w^2}\right)}}{w^2}$$

The particle acceleration is

$$\frac{\partial^2 y}{\partial t^2} = v^2\frac{d^2y}{dy^2} = -\frac{2v^2\,Ae^{\left(-\frac{z^2}{w^2}\right)}}{w^2} + \frac{4v^2\,Az^2e^{\left(-\frac{z^2}{w^2}\right)}}{w^4}$$

The slope of the curve is

$$\frac{\partial y}{\partial x} = \frac{dy}{dz} = -\frac{2\,Aze^{\left(-\frac{z^2}{w^2}\right)}}{w^2}$$

and the curvature is given by

$$\frac{\partial^2 y}{\partial x^2} = \frac{d^2y}{dz^2} = -\frac{2\,Ae^{\left(-\frac{z^2}{w^2}\right)}}{w^2} + \frac{4\,Az^2e^{\left(-\frac{z^2}{w^2}\right)}}{w^4}$$

2.4 *Harmonic waves*

Angular frequency and wave number

We turn our attention now to waves of the form[3]

$$y(x,t) = A\cos[k(x-vt)+\alpha] \tag{2.15}$$

As we did with the simple harmonic oscillator before, we begin by exploring the physical meaning of the constants A, k, and α.

First, let's look at the motion of one particle, at position $x = x_0$. Its motion is that of a simple harmonic oscillator. Rearranging the terms in the phase a bit, we can write it as

$$y(x_0,t) = A\cos(-kvt + kx_0 + \alpha) \tag{2.16}$$

And purely for esthetic reasons, let's avail ourselves of the fact that cosine is an even function, so we can change the sign of its argument:

$$y(x_0,t) = A\cos(kvt - kx_0 - \alpha) \tag{2.17}$$

Now this looks exactly like the familiar SHO, if we identify kv with the angular frequency ω, A with the amplitude of the oscillation, and the constant terms $-(kx_0+\alpha)$ with the initial phase ϕ_0. So we can go back to the expression for a harmonic wave, and rewrite it as

$$y(x,t) = A\cos(kx - \omega t + \alpha) \tag{2.18}$$

[3] To avoid muddling the math in what follows, we assume the wave is propagating in the $+x$ direction. If it was propagating in the opposite direction, a sign change would be required in the argument of the cosine function. We will consider this later.

We still want to understand what the constant k is. To do this, consider now a snapshot of the wave at an instant $t = t_0$:

$$y(x, t_0) = A\cos(kx - \omega t_0 + \alpha) \tag{2.19}$$

This is a periodic function of position x, and it repeats every time the argument of the cosine function advances by 2π:

$$\begin{aligned} A\cos(kx - \omega t_0 + \alpha) &= A\cos(kx + \mathbf{2\pi} - \omega t_0 + \alpha) = \\ &= A\cos[k(x + 2\pi/k) - \omega t_0 + \alpha] \end{aligned} \tag{2.20}$$

Another way of looking at it is that the function repeats when x advances by $2\pi/k$. This has dimensions of length, and we call it the **wavelength** of the wave, λ. Then

$$k = \frac{2\pi}{\lambda} \tag{2.21}$$

If we compare this with $\omega = 2\pi/T$, we see that k plays the same role as ω, but in terms of the "spatial period" λ instead of the "temporal period" T; k is a "spatial angular frequency". It is actually not very helpfully called the **wave number**. Dimensionally, it is an inverse length.

The speed of the wave

If we compare the expression of equation (2.15) and that of equation (2.18), we find the following important relation between the three quantities v, ω, and k:

$$v = \frac{\omega}{k} \tag{2.22}$$

In terms of f and λ,

$$v = f\lambda \tag{2.23}$$

Example 2.2: *Figure 2.6 shows the oscillation at $x = 5$ cm produced by a wave of the form*

$$y(x, t) = A\cos(\omega t - kx + \phi_0)$$

The speed of the wave is 5.0 cm/s. Calculate the amplitude, the angular frequency, the wavelength, and the initial phase of the wave.

The wave oscillates between ± 0.50 cm, so $A = 0.5$ cm.
The period is 2.0 s, so $\omega = \pi$ rad/s.
Since $v = \lambda/T$, the wavelength is $\lambda = 10$ cm.
At $t = 0$, $y = A\cos(-kx + \phi_0)$, and $y = 0.25$ cm, so

$$\phi_0 = kx + \arccos(y/A) = \pi \pm 1.047\,\text{rad} \tag{2.24}$$

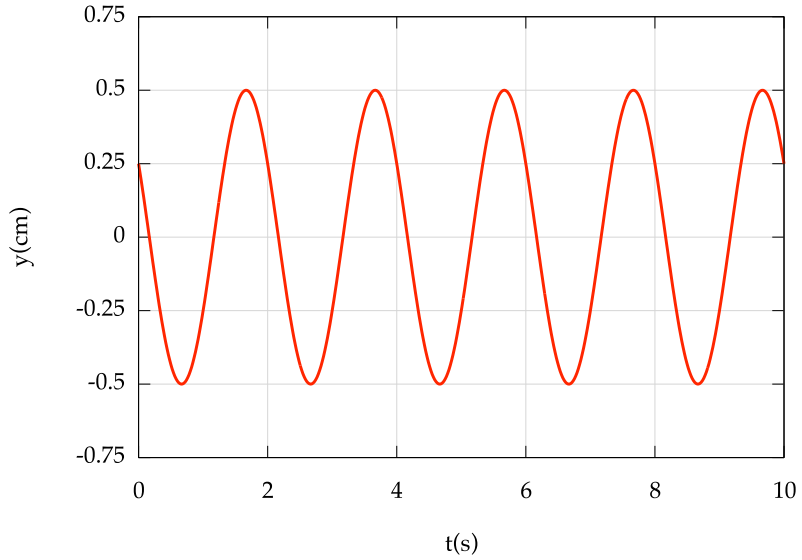


Figure 2.6: Time evolution of a wave at $x = 5$ cm.

since there are two possible values of $\arccos(y/A)$. To decide which one is correct, we calculate the particle velocity at $t = 0$:

$$\frac{\partial y}{\partial t} = -A\sin(kx + \phi_0)$$

From the graph we see that the particle velocity is negative, so $\sin(kx + \phi_0) > 0$, which is satisfied by $\phi_0 = \pi + 1.047 = 4.189$ rad.

Equivalent ways of expressing a simple harmonic wave

Equation (2.18) is the standard representation of a harmonic wave travelling in the $+x$ direction. If it is propagating in the $-x$ direction, then the expression becomes

$$y(x,t) = A\cos(kx + \omega t + \alpha) \tag{2.25}$$

Finally, just like with simple harmonic oscillations, a harmonic wave can also be written in terms of a sine function instead of a cosine function, if so desired:

$$y(x,t) = A\sin(kx - \omega t + \beta) \tag{2.26}$$

The phase constant β is just different.

Also, kx and ωt can be exchanged:

$$y(x,t) = A\cos(\omega t - kx + \gamma) \tag{2.27}$$

again provided we change the phase constant appropriately.

Example 2.3: *A wave is represented as in equation (2.18). If we want to represent it instead in the form of equation (2.26) or equation (2.27), what are the values of β and γ in terms of α?*

We can set $x = 0$ and $t = 0$ for simplicity, so in the first case we are left with the condition

$$\cos(\alpha) = \sin(\beta)$$

This means that $\beta = \pi/2 - \alpha$.

In the second case, since the cosine function is an even function,

$$\cos(\omega t - kx + \gamma) = \cos(kx - \omega t - \gamma)$$

so $\gamma = -\alpha$.

2.5 *Complex representation of harmonic waves*

We saw before that a simple harmonic oscillation in the form of a cosine function (or a sine function) could be regarded as the real (or imaginary) part of a complex exponential. And the problem of superposing two or more oscillations was made computationally simpler by adding the complex oscillations, or phasors, and then taking the real (or imaginary) part of the result at the end.

Superposition of waves is also handled more conveniently if we treat a harmonic wave as the real or the imaginary part of a complex wave

$$\tilde{y}(x,t) = Ae^{i(kx-\omega t+\alpha)} \tag{2.28}$$

or

$$\tilde{y}(x,t) = \tilde{A}e^{i(kx-\omega t)} \tag{2.29}$$

where the complex amplitude $\tilde{A} = Ae^{i\alpha}$ includes the phase constant α.

2.6 *Waves in 2 and 3 dimensions*

So far we have studied only waves which propagate in 1 dimension, along a line. However, our analysis also extends easily to waves that propagate in 2 and 3 dimensions but that due to their symmetry, can be described mathematically using only one coordinate.

A 2D example is a wave generated by a point source on a liquid surface, propagating outwards uniformly in all directions. An instance



Figure 2.7: "Water spire" by likeablerodent

of this is shown in Figure 2.6. This wave can still be represented as a cosine function, where the spatial coordinate is r, a radial coordinate from the source, which is at the centre of the ripple pattern. If we call y the height of the water surface above the equilibrium height of the still pond, at a distance r from the source, we can represent y by

$$y(r,t) = A\cos(kr - \omega t) \tag{2.30}$$

The ripples look like circles because if we look at the peaks, for example, they correspond to points where y is maximum, i.e. $kr - \omega t = 2n\pi$, where n is a whole number. This is equivalent to

$$r = \frac{2n\pi + \omega t}{k} \tag{2.31}$$

At any given time t, r is a constant, which means that the points on a crest of the wave lie on a circle. Some jargon: The locus of all points with the same phase, or argument of the cosine function, is called a **wavefront**. The wave shown in Figure 2.6 has circular wavefronts.

A similar situation can arise in 3 dimensions, when a point source generates a wave that spreads equally in all directions. In that case, the wave is still represented by equation (2.30), but the wavefronts are now spherical.

If the source is a line rather than a point, the resulting wave is yet again represented by equation (2.30), but r takes on a different meaning: it is the perpendicular distance from the line source. The wavefronts are cylindrical.

If the source is a plane (think for example of a vibrating membrane), then the wavefronts are parallel planes, and we can still represent the wave with the familiar $y(x,t) = A\cos(kx - \omega t)$, where x is the perpendicular distance from the plane source. Figure 2.6 shows examples of cylindrical and plane waves.

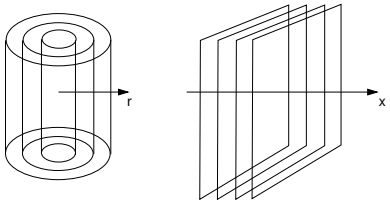


Figure 2.8: Cylindrical and plane wavefronts.